NTA JEE Mains Jan 2026

Test Date	28/01/2026
Test Time	9:00 AM - 12:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 Let f be a polynomial function such that $f(x^2+1) = x^4 + 5x^2 + 2$, for all $x \in \mathbb{R}$.

Then $\int_0^3 f(x)dx$ is equal to

Options
1. $\frac{41}{3}$
2. $\frac{27}{2}$
3. $\frac{33}{2}$
4. $\frac{5}{3}$

Question Type : **MCQ**
Question ID : **444792691**
Option 1 ID : **4447922358**
Option 2 ID : **4447922356**
Option 3 ID : **4447922357**
Option 4 ID : **4447922359**

Q.2 For three unit vectors $\vec{a}, \vec{b}, \vec{c}$ satisfying

$$\left|\vec{a}-\vec{b}\right|^2 + \left|\vec{b}-\vec{c}\right|^2 + \left|\vec{c}-\vec{a}\right|^2 = 9 \text{ and } \left|2\vec{a}+k\vec{b}+k\vec{c}\right| = 3,$$

the positive value of k is

Options
1. 4
2. 5
3. 3
4. 6

Question Type : **MCQ**
Question ID : **444792689**
Option 1 ID : **4447922349**
Option 2 ID : **4447922350**
Option 3 ID : **4447922348**
Option 4 ID : **4447922351**

Q.3 If the distances of the point $(1, 2, a)$ from the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$ along the lines $L_1: \frac{x-1}{3}=\frac{y-2}{4}=\frac{z-a}{b}$ and $L_2: \frac{x-1}{1}=\frac{y-2}{4}=\frac{z-a}{c}$ are equal, then $a+b+c$ is equal to

Options
1. 6
2. 4
3. 5
4. 7

Question Type : **MCQ**
Question ID : **444792688**
Option 1 ID : **4447922345**
Option 2 ID : **4447922347**
Option 3 ID : **4447922346**
Option 4 ID : **4447922344**

Q.4 The area of the region $R = \{(x, y) : xy \le 8,\ 1 \le y \le x^2,\ x \ge 0\}$ is

Options
1. $\frac{1}{3}\left(40\log_e(2)+27\right)$
2. $\frac{2}{3}\left(24\log_e(2)-7\right)$
3. $\frac{2}{3}\left(20\log_e(2)+9\right)$
4. $\frac{1}{3}\left(49\log_e(2)-15\right)$

Question Type : **MCQ**
Question ID : **444792692**
Option 1 ID : **4447922363**
Option 2 ID : **4447922361**
Option 3 ID : **4447922360**
Option 4 ID : **4447922362**

Q.5 The value of $\sum_{k=1}^{\infty}(-1)^{k+1}\left(\frac{k(k+1)}{k!}\right)$ is

Options
1. $1/e$
2. $2/e$
3. $\sqrt{e}$
4. $e/2$

Question Type : **MCQ**
Question ID : **444792690**
Option 1 ID : **4447922354**
Option 2 ID : **4447922353**
Option 3 ID : **4447922355**
Option 4 ID : **4447922352**

Q.6 If $g(x) = 3x^2 + 2x - 3$, $f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:

Options

1. $\frac{7}{2}$
2. $-\frac{25}{6}$
3. $\frac{25}{6}$
4. $-\frac{7}{2}$

Question Type : **MCQ**
Question ID : **444792677**
Option 1 ID : **4447922302**
Option 2 ID : **4447922303**
Option 3 ID : **4447922300**
Option 4 ID : **4447922301**

Q.7 The common difference of the A.P.: $a_1, a_2, \ldots, a_m$ is 13 more than the common difference of the A.P.: $b_1, b_2, \ldots, b_n$. If $b_{31} = -277$, $b_{43} = -385$ and $a_{78} = 327$, then a_1 is equal to

Options

1. 19
2. 24
3. 16
4. 21

Question Type : **MCQ**
Question ID : **444792681**
Option 1 ID : **4447922317**
Option 2 ID : **4447922319**
Option 3 ID : **4447922316**
Option 4 ID : **4447922318**

Q.8 Let $y = y(x)$ be the solution of the differential equation

$$x\frac{dy}{dx} - \sin 2y = x^3\left(2 - x^3\right)\cos^2 y, \quad x \neq 0.$$

If $y(2) = 0$, then $\tan(y(1))$ is equal to

Options

1. $-\frac{7}{4}$
2. $\frac{7}{4}$
3. $-\frac{3}{4}$
4. $\frac{3}{4}$

Question Type : **MCQ**
Question ID : **444792694**
Option 1 ID : **4447922368**
Option 2 ID : **4447922370**
Option 3 ID : **4447922369**
Option 4 ID : **4447922371**

Q.9 Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x+2\sqrt{2}\,y=4$. If the co-ordinates of the vertex A are (α, β), then the greatest integer less than or equal to $\left|\alpha+\sqrt{2}\beta\right|$ is

Options
1. 4
2. 2
3. 3
4. 5

Question Type : **MCQ**
Question ID : **444792685**
Option 1 ID : **4447922334**
Option 2 ID : **4447922332**
Option 3 ID : **4447922333**
Option 4 ID : **4447922335**

Q.10 If $\dfrac{\tan(A-B)}{\tan A}+\dfrac{\sin^2 C}{\sin^2 A}=1,\ A, B, C \in\left(0, \dfrac{\pi}{2}\right)$, then

Options
1. tan A, tan B, tan C are in G.P.
2. tan A, tan C, tan B are in G.P.
3. tan A, tan B, tan C are in A.P.
4. tan A, tan C, tan B are in A.P.

Question Type : **MCQ**
Question ID : **444792687**
Option 1 ID : **4447922340**
Option 2 ID : **4447922342**
Option 3 ID : **4447922341**
Option 4 ID : **4447922343**

Q.11 The value of

$$\lim_{x\to 0} \frac{\log_e\left(\sec(ex)\cdot\sec\left(e^2x\right)\cdot\ldots\cdot\sec\left(e^{10}x\right)\right)}{e^2 - e^{2\cos x}}$$

is equal to

Options

1. $\dfrac{\left(e^{20}-1\right)}{2e^2\left(e^2-1\right)}$
2. $\dfrac{\left(e^{20}-1\right)}{2\left(e^2-1\right)}$
3. $\dfrac{\left(e^{10}-1\right)}{2\left(e^2-1\right)}$
4. $\dfrac{\left(e^{10}-1\right)}{2e^2\left(e^2-1\right)}$

Question Type : **MCQ**
Question ID : **444792693**
Option 1 ID : **4447922365**
Option 2 ID : **4447922364**
Option 3 ID : **4447922366**
Option 4 ID : **4447922367**

Q.12 Let z be a complex number such that $|z-6|=5$ and $|z+2-6i|=5$. Then the value of $z^3+3z^2-15z+141$ is equal to

Options

1. 50
2. 37
3. 42
4. 61

Question Type : **MCQ**
Question ID : **444792679**
Option 1 ID : **4447922310**
Option 2 ID : **4447922308**
Option 3 ID : **4447922309**
Option 4 ID : **4447922311**

Q.13 If α, β, where $\alpha < \beta$, are the roots of the equation $\lambda x^2 - (\lambda + 3)x + 3 = 0$ such that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$, then the sum of all possible values of λ is

Options
1. 4
2. 6
3. 2
4. 8

Question Type : **MCQ**
Question ID : **444792678**
Option 1 ID : **4447922305**
Option 2 ID : **4447922306**
Option 3 ID : **4447922304**
Option 4 ID : **4447922307**

Q.14 Let S = {1, 2, 3, 4, 5, 6, 7, 8, 9}. Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9-digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then,

Options
1. $29x = 5y$
2. $21x = 4y$
3. $56x = 9y$
4. $45x = 7y$

Question Type : **MCQ**
Question ID : **444792684**
Option 1 ID : **4447922331**
Option 2 ID : **4447922328**
Option 3 ID : **4447922330**
Option 4 ID : **4447922329**

Q.15 A bag contains 10 balls out of which k are red and $(10 - k)$ are black, where $0 \leq k \leq 10$. If three balls are drawn at random without replacement and all of them are found to be black, then the probability that the bag contains 1 red and 9 black balls is:

Options
1. $\frac{7}{110}$
2. $\frac{7}{11}$
3. $\frac{14}{55}$
4. $\frac{7}{55}$

Question Type : **MCQ**
Question ID : **444792682**
Option 1 ID : **4447922323**
Option 2 ID : **4447922320**
Option 3 ID : **4447922321**
Option 4 ID : **4447922322**

Q.16 Let A, B and C be three 2×2 matrices with real entries such that $B = (I + A)^{-1}$ and $A + C = I$. If $BC = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$ and $CB\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$, then $x_1 + x_2$ is

Options
1. 2
2. 0
3. -2
4. 4

Question Type : **MCQ**
Question ID : **444792680**
Option 1 ID : **4447922313**
Option 2 ID : **4447922312**
Option 3 ID : **4447922314**
Option 4 ID : **4447922315**

Q.17 Let $y = x$ be the equation of a chord of the circle C_1 (in the closed half-plane $x \geq 0$) of diameter 10 passing through the origin. Let C_2 be another circle described on the given chord as its diameter. If the equation of the chord of the circle C_2, which passes through the point (2, 3) and is farthest from the center of C_2, is $x + ay + b = 0$, then $a - b$ is equal to

Options
1. -2
2. -6
3. 6
4. 10

Question Type : **MCQ**
Question ID : **444792686**
Option 1 ID : **4447922337**
Option 2 ID : **4447922336**
Option 3 ID : **4447922338**
Option 4 ID : **4447922339**

Q.18 The mean and variance of 10 observations are 9 and 34.2, respectively. If 8 of these observations are 2, 3, 5, 10, 11, 13, 15, 21, then the mean deviation about the median of all the 10 observations is

Options
1. 4
2. 5
3. 7
4. 6

Question Type : **MCQ**
Question ID : **444792683**
Option 1 ID : **4447922327**
Option 2 ID : **4447922326**
Option 3 ID : **4447922324**
Option 4 ID : **4447922325**

Q.19 Let $S = \{x^3 + ax^2 + bx + c : a, b, c \in \mathbb{N} \text{ and } a, b, c \le 20\}$ be a set of polynomials. Then the number of polynomials in S, which are divisible by $x^2 + 2$, is

Options
1. 20
2. 6
3. 10
4. 120

Question Type : **MCQ**
Question ID : **444792676**
Option 1 ID : **4447922298**
Option 2 ID : **4447922296**
Option 3 ID : **4447922297**
Option 4 ID : **4447922299**

Q.20 If $\int \left(\frac{1 - 5\cos^2 x}{\sin^5 x \cos^2 x} \right) dx = f(x) + C$, where C is the constant of integration, then $f\left(\frac{\pi}{6}\right) - f\left(\frac{\pi}{4}\right)$ is equal to

Options
1. $\frac{1}{\sqrt{3}}\left(26 + \sqrt{3}\right)$
2. $\frac{4}{\sqrt{3}}\left(8 - \sqrt{6}\right)$
3. $\frac{1}{\sqrt{3}}\left(26 - \sqrt{3}\right)$
4. $\frac{2}{\sqrt{3}}\left(4 + \sqrt{6}\right)$

Question Type : **MCQ**
Question ID : **444792695**
Option 1 ID : **4447922372**
Option 2 ID : **4447922373**
Option 3 ID : **4447922375**
Option 4 ID : **4447922374**

Section : **Mathematics Section B**

Q.21 For some $\theta \in \left(0, \frac{\pi}{2}\right)$, let the eccentricity and the length of the latus rectum of the hyperbola $x^2 - y^2 \sec^2\theta = 8$ be e_1 and l_1, respectively, and let the eccentricity and the length of the latus rectum of the ellipse $x^2 \sec^2\theta + y^2 = 6$ be e_2 and l_2, respectively. If $e_1^2 = e_2^2\left(\sec^2\theta + 1\right)$, then $\left(\frac{l_1 l_2}{e_1 e_2}\right)\tan^2\theta$ is equal to ______

Question Type : **SA**
Question ID : **444792700**

Q.22 Let PQR be a triangle such that $\overrightarrow{PQ} = -2\hat{i} - \hat{j} + 2\hat{k}$ and $\overrightarrow{PR} = a\hat{i} + b\hat{j} - 4\hat{k}$, $a, b \in \mathbb{Z}$. Let S be the point on QR, which is equidistant from the lines PQ and PR. If $|\overrightarrow{PR}| = 9$ and $\overrightarrow{PS} = \hat{i} - 7\hat{j} + 2\hat{k}$, then the value of $3a - 4b$ is _____

Question Type : **SA**
Question ID : **444792698**

Q.23 The value of $\sum_{r=1}^{20}\left(\left|\sqrt{\pi\left(\int_0^r x|\sin \pi x|\,dx\right)}\right|\right)$ is________

Question Type : **SA**
Question ID : **444792699**

Q.24 If $k = \tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\left(\frac{2}{3}\right)\right) + \tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2}{3}\right)\right)$, then the number of solutions of the equation $\sin^{-1}(kx - 1) = \sin^{-1} x - \cos^{-1} x$ is _________

Question Type : **SA**
Question ID : **444792697**

Q.25 In a G.P., if the product of the first three terms is 27 and the set of all possible values for the sum of its first three terms is $\mathbb{R} - (a, b)$, then $a^2 + b^2$ is equal to _______.

Question Type : **SA**
Question ID : **444792696**

Section : **Physics Section A**

Q.26 Given below are two statements:

Statement I: A plane wave after passing through prism remains as plane wave but passing through small pin hole may become spherical wave.

Statement II: The curvature of a spherical wave emerging from a slit will increase for increasing slit width.

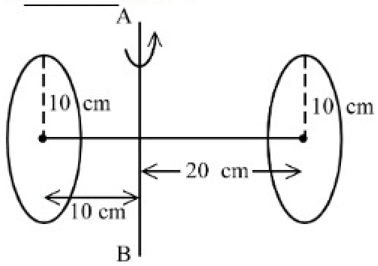
In the light of the above statements, choose the ***correct*** answer from the options given below

Options
1. Both Statement I and Statement II are true
2. Statement I is false but Statement II is true
3. Statement I is true but Statement II is false
4. Both Statement I and Statement II are false

Question Type : **MCQ**
Question ID : **444792717**
Option 1 ID : **4447922445**
Option 2 ID : **4447922448**
Option 3 ID : **4447922447**
Option 4 ID : **4447922446**

Q.27 Two circular discs of radius each 10 cm are joined at their centres by a rod of length 30 cm and mass 600 gm as shown in figure.

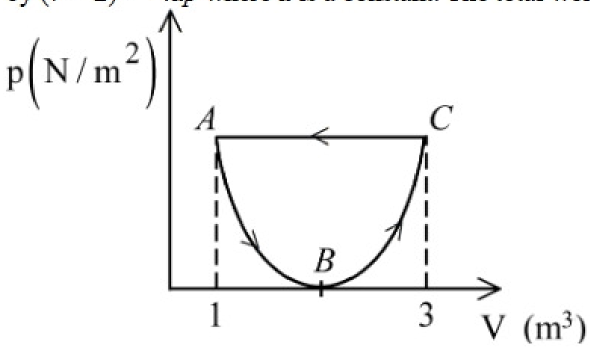
If the mass of each disc is 600 gm and applied torque between two discs is 43×10^5 dyne.cm, the angular acceleration of the discs about the given axis AB is_______rad/s^2.



Options
1. 22
2. 100
3. 11
4. 27

Question Type : **MCQ**
Question ID : **444792706**
Option 1 ID : **4447922402**
Option 2 ID : **4447922404**
Option 3 ID : **4447922403**
Option 4 ID : **4447922401**

Q.28 In the following p-V diagram the equation of state along the curved path is given by $(V-2)^2 = 4ap$ where a is a constant. The total work done in the closed path is

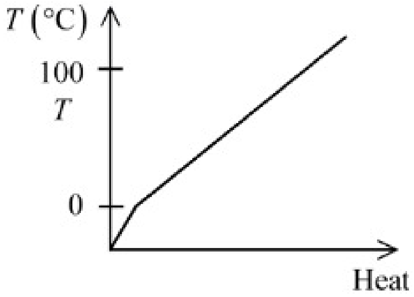


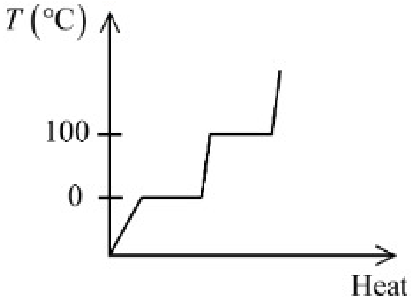
Options
1. $+\frac{1}{3a}$
2. $\frac{1}{2a}$
3. $-\frac{1}{3a}$
4. $-\frac{1}{a}$

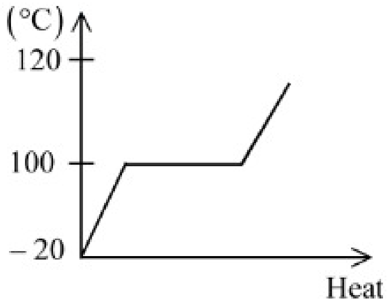
Question Type : **MCQ**
Question ID : **444792709**
Option 1 ID : **4447922415**
Option 2 ID : **4447922416**
Option 3 ID : **4447922414**
Option 4 ID : **4447922413**

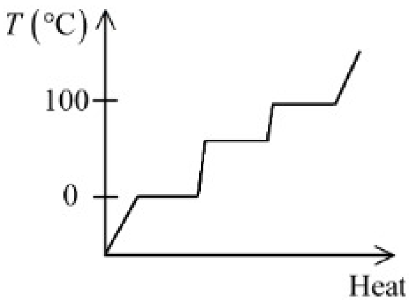
Q.29 Which of the following best represents the temperature versus heat supplied graph for water, in the range of −20 °C to 120 °C ?

Options

1. 

2. 

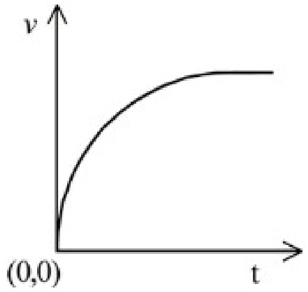
3. 

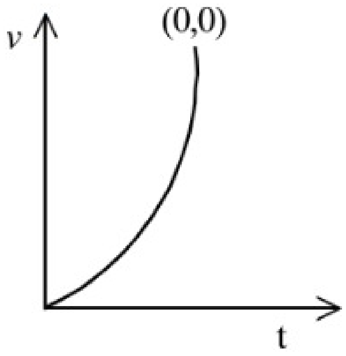
4. 

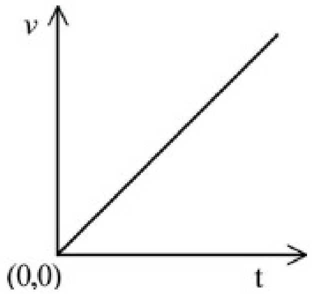
Question Type :	**MCQ**
Question ID :	**444792710**
Option 1 ID :	**4447922418**
Option 2 ID :	**4447922417**
Option 3 ID :	**4447922419**
Option 4 ID :	**4447922420**

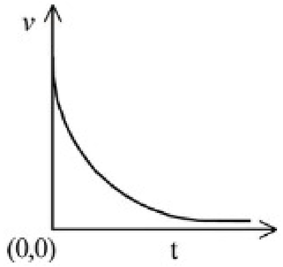
Q.30 A particle of mass m falls from rest through a resistive medium having resistive force, $F = -kv$, where v is the velocity of the particle and k is a constant. Which of the following graphs represents velocity (v) versus time (t)?

Options

1. 

2. 

3. 

4. 

Question Type : **MCQ**
Question ID : **444792703**
Option 1 ID : **4447922390**
Option 2 ID : **4447922392**
Option 3 ID : **4447922391**
Option 4 ID : **4447922389**

Q.31 For the two cells having same EMF E and internal resistance r, the current passing through the external resistor 6 Ω is same when both the cells are connected either in parallel or in series. The value of internal resistance r is ________Ω.

Options
1. 6
2. 4
3. 3
4. 9

Question Type : **MCQ**
Question ID : **444792712**
Option 1 ID : **4447922427**
Option 2 ID : **4447922426**
Option 3 ID : **4447922425**
Option 4 ID : **4447922428**

Q.32 The electric field of an electromagnetic wave travelling through a medium is given by $\vec{E}(x,t)=25\sin(2.0\times10^{15}t-10^{7}\,x)\hat{n}$
then the refractive index of the medium is _________.
(All given measurement are in SI units)

Options
1. 1.2
2. 1.5
3. 2
4. 1.7

Question Type : **MCQ**
Question ID : **444792716**
Option 1 ID : **4447922442**
Option 2 ID : **4447922441**
Option 3 ID : **4447922443**
Option 4 ID : **4447922444**

Q.33 In the potentiometer, when the cell in the secondary circuit is shunted with 4 Ω resistance, the balance is obtained at the length 120 cm of wire. Now when the same cell is shunted with 12 Ω resistance, the balance is shifted to a length of 180 cm. The internal resistance of cell is _______Ω

Options
1. 6
2. 12
3. 3
4. 4

Question Type : **MCQ**
Question ID : **444792702**
Option 1 ID : **4447922387**
Option 2 ID : **4447922388**
Option 3 ID : **4447922385**
Option 4 ID : **4447922386**

Q.34 Water drops fall from a tap on the floor, 5 m below, at regular intervals of time, the first drop strikes the floor when the sixth drop begins to fall. The height at which the fourth drop will be from ground, at the instant when the first drop strikes the ground is __________m.

($g = 10\ m/s^2$)

Options
1. 3.8
2. 4.0
3. 4.2
4. 2.5

Question Type : **MCQ**
Question ID : **444792705**
Option 1 ID : **4447922400**
Option 2 ID : **4447922397**
Option 3 ID : **4447922399**
Option 4 ID : **4447922398**

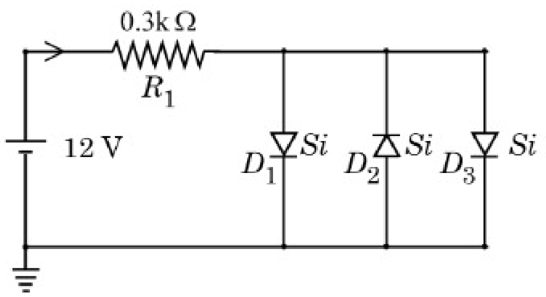
Q.35 The magnetic field at the centre of a current carrying circular loop of radius R is 16 μT. The magnetic field at a distance $x = \sqrt{3}R$ on its axis from the centre is ______μT.

Options
1. $2\sqrt{2}$
2. 2
3. 4
4. 8

Question Type : **MCQ**
Question ID : **444792711**
Option 1 ID : **4447922422**
Option 2 ID : **4447922421**
Option 3 ID : **4447922423**
Option 4 ID : **4447922424**

Q.36 Assuming in forward bias condition there is a voltage drop of 0.7 V across a silicon diode, the current through diode D_1 in the circuit is ______mA.

(Assume all diodes in the given circuit are identical)



Options
1. 11.7
2. 20.15
3. 18.8
4. 17.6

Question Type : **MCQ**
Question ID : **444792720**
Option 1 ID : **4447922459**
Option 2 ID : **4447922458**
Option 3 ID : **4447922460**
Option 4 ID : **4447922457**

Q.37 A block of mass 5 kg is moving on an inclined plane which makes an angle of $30°$ with the horizontal. Friction coefficient between the block and inclined plane surface is $\frac{\sqrt{3}}{2}$. The force to be applied on the block so that the block will move down without acceleration is _______N.
($g = 10\ \text{m/s}^2$) .

Options
1. 15
2. 25
3. 12.5
4. 7.5

Question Type : **MCQ**
Question ID : **444792704**
Option 1 ID : **4447922396**
Option 2 ID : **4447922395**
Option 3 ID : **4447922393**
Option 4 ID : **4447922394**

Q.38 When both jaws of vernier callipers touch each other, zero mark of the vernier scale is right to zero mark of main scale, 4^{th} mark on vernier scale coincides with certain mark on the main scale. While measuring the length of a cylinder, observer observes 15 divisions on main scale and 5^{th} division of vernier scale coincides with a main scale division. Measured length of cylinder is ________mm.
(Least count of Vernier calliper = 0.1 mm)

Options
1. 15.5
2. 15.4
3. 15.9
4. 15.1

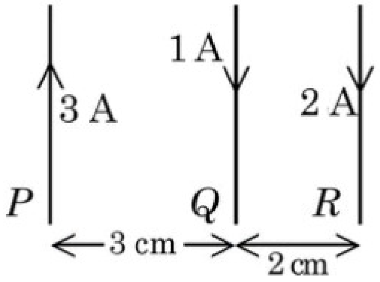
Question Type : **MCQ**
Question ID : **444792701**
Option 1 ID : **4447922382**
Option 2 ID : **4447922384**
Option 3 ID : **4447922381**
Option 4 ID : **4447922383**

Q.39 The electric current in the circuit is given as $i = i_0(t/T)$. The r.m.s current for the period $t = 0$ to $t = T$ is_______.

Options
1. $\frac{i_0}{\sqrt{2}}$
2. $\frac{i_0}{\sqrt{3}}$
3. $\frac{i_0}{\sqrt{6}}$
4. i_0

Question Type : **MCQ**
Question ID : **444792713**
Option 1 ID : **4447922430**
Option 2 ID : **4447922431**
Option 3 ID : **4447922432**
Option 4 ID : **4447922429**

Q.40 Three long straight wires carrying current are arranged mutually parallel as shown in the figure. The force experienced by 15 cm length of wire Q is________.



($\mu_0 = 4\pi \times 10^{-7}$ T.m/A)

Options
1. 6×10^{-7} N towards R
2. 6×10^{-7} N towards P
3. 6×10^{-6} N towards R
4. 6×10^{-6} N towards P

Question Type : **MCQ**
Question ID : **444792714**
Option 1 ID : **4447922433**
Option 2 ID : **4447922434**
Option 3 ID : **4447922436**
Option 4 ID : **4447922435**

Q.41 Two wires A and B made of different materials of lengths 6.0 cm and 5.4 cm, respectively and area of cross sections 3.0×10^{-5} m^2 and 4.5×10^{-5} m^2, respectively are stretched by the same magnitude under a given load. The ratio of the Young's modulus of A to that of B is x : 3. The value of x is______.

Options
1. 2
2. 4
3. 5
4. 1

Question Type : **MCQ**
Question ID : **444792707**
Option 1 ID : **4447922406**
Option 2 ID : **4447922407**
Option 3 ID : **4447922408**
Option 4 ID : **4447922405**

Q.42 The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a plano-concave lens (refractive index = 1.7) are same. If the curvature of plano-concave lens exactly matches with the curvature of back surface of the biconvex lens, then ratio of radius of curvature of front and back surface of the biconvex lens is_______.

Options
1. 12 : 5
2. 5 : 2
3. 2 : 5
4. 5 : 12

Question Type : **MCQ**
Question ID : **444792718**
Option 1 ID : **4447922452**
Option 2 ID : **4447922449**
Option 3 ID : **4447922451**
Option 4 ID : **4447922450**

Q.43 An atom ${}^{8}_{3}X$ is bombarded by shower of fundamental particles and in 10 s this atom absorbed 10 electrons, 10 protons and 9 neutrons. The percentage growth in the surface area of the nucleons is recorded by:

Options
1. 225%
2. 250%
3. 150 %
4. 900%

Question Type : **MCQ**
Question ID : **444792719**
Option 1 ID : **4447922455**
Option 2 ID : **4447922456**
Option 3 ID : **4447922453**
Option 4 ID : **4447922454**

Q.44 Two point charges of 1 nC and 2 nC are placed at the two corners of equilateral triangle of side 3 cm. The work done in bringing a charge of 3 nC from infinity to the third corner of the triangle is _____ μJ.

$$\frac{1}{4\pi \epsilon_0} = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$$

Options
1. 2.7
2. 3.3
3. 27
4. 5.4

Question Type : **MCQ**
Question ID : **444792715**
Option 1 ID : **4447922440**
Option 2 ID : **4447922437**
Option 3 ID : **4447922439**
Option 4 ID : **4447922438**

Q.45 10 kg of ice at –10 °C is added to 100 kg of water to lower its temperature from 25 °C. Consider no heat exchange to surroundings. The decrement to the temperature of water is __________°C.
(specific heat of ice = 2100 J/Kg.°C, specific heat of water = 4200 J/Kg.°C, latent heat of fusion of ice = 3.36×10^5 J/Kg)

Options
1. 11.6
2. 10
3. 6.67
4. 15

Question Type : **MCQ**
Question ID : **444792708**
Option 1 ID : **4447922409**
Option 2 ID : **4447922412**
Option 3 ID : **4447922411**
Option 4 ID : **4447922410**

Section : **Physics Section B**

Q.46 A convex lens of refractive index 1.5 and focal length f = 18 cm is immersed in water. The difference in focal lengths of the given lens when it is in water and in air is $\alpha \times$ f. The value of α is _____.
(refractive index of water = 4/3)

Question Type : **SA**
Question ID : **444792721**

Q.47 A solid sphere of radius 10 cm is rotating about an axis which is at a distance 15 cm from its centre. The radius of gyration about this axis is $\sqrt{n}$ cm. The value of n is

Question Type : **SA**
Question ID : **444792724**

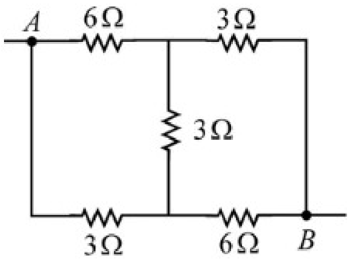
Q.48 The ratio of de Broglie wavelength of a deutron with kinetic energy E to that of an alpha particle with kinetic energy $2E$, is $n : 1$. The value of n is ____.
(Assume mass of proton = mass of neutron) :

Question Type : **SA**
Question ID : **444792723**

Q.49 The displacement of a particle, executing simple harmonic motion with time period T, is expressed as $x(t) = A \sin \omega t$, where A is the amplitude. The maximum value of potential energy of this oscillator is found at $t = T/2\beta$. The value of β is ______.

Question Type : **SA**
Question ID : **444792725**

Q.50 The equivalent resistance between the points A and B in the following circuit is $\frac{x}{5}\Omega$. The value of x is __________.



Question Type : **SA**
Question ID : **444792722**

Section : **Chemistry Section A**

Q.51 Given below are two statements:

Statement I: The number of species among BF_4^-, SiF_4, XeF_4 and SF_4, that have unequal E–F bond lengths is two. Here, E is the central atom.

Statement II: Among O_2^-, O_2^{2-}, F_2 and O_2^+, O_2^- has the highest bond order.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options
1. Statement I is false but Statement II is true
2. Both Statement I and Statement II are true
3. Both Statement I and Statement II are false
4. Statement I is true but Statement II is false

Question Type : **MCQ**
Question ID : **444792728**
Option 1 ID : **4447922477**
Option 2 ID : **4447922474**
Option 3 ID : **4447922475**
Option 4 ID : **4447922476**

Q.52 20.0 dm^3 of an ideal gas 'X' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until pressure of the gas is 0.2 MPa. Which of the following option is correct?

(Given: log 2 = 0.3010 and log 5 = 0.6989)

Options
1. $w = 9.1\ J, \Delta U = 9.1\ J, \Delta H = 0; q = 0$
2. $w = -3.9\ kJ, \Delta U = 0, \Delta H = 0; q = 3.9\ kJ$
3. $w = +4.1\ kJ, \Delta U = 0\ , \Delta H = 0; q = -4.1\ kJ$
4. $w = -9.1\ kJ, \Delta U = 0, \Delta H = 0, q = 9.1\ kJ$

Question Type : **MCQ**
Question ID : **444792729**
Option 1 ID : **4447922479**
Option 2 ID : **4447922478**
Option 3 ID : **4447922481**
Option 4 ID : **4447922480**

Q.53 An organic compound undergoes first order decomposition. The time taken for decomposition to $\left(\frac{1}{8}\right)^{th}$ and $\left(\frac{1}{10}\right)^{th}$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively.

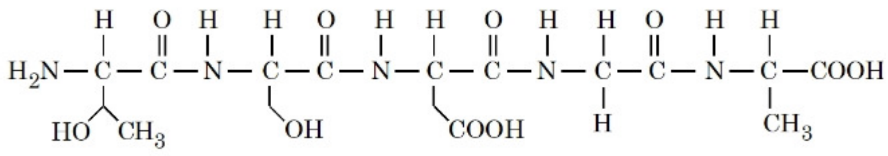
What is the value of $\frac{t_{1/8}}{t_{1/10}} \times 10$?

(log 2= 0.3)

Options
1. 30
2. 9
3. 3
4. 0.9

Question Type : **MCQ**
Question ID : **444792732**
Option 1 ID : **4447922492**
Option 2 ID : **4447922491**
Option 3 ID : **4447922490**
Option 4 ID : **4447922493**

Q.54 In the given pentapeptide, find out an essential amino acid (Y) and the sequence present in the pentapeptide:



Choose the correct answer from the options given below:

Options

1.

(Y)	(Sequence)
Serine	Ser−Asp−Thr−Ala−Gly

2.

(Y)	(Sequence)
Threonine	Thr−Ser−Asp−Gly−Ala

3.

(Y)	(Sequence)
Serine	Thr−Ser−Asp−Ala−Gly

4.

(Y)	(Sequence)
Threonine	Ser−Thr−Asp−Gly−Ala

Question Type : **MCQ**
Question ID : **444792744**
Option 1 ID : **4447922540**
Option 2 ID : **4447922538**
Option 3 ID : **4447922539**
Option 4 ID : **4447922541**

Q.55 Given below are two statements:

Statement I: The number of pairs, from the following, in which both the ions are coloured in aqueous solution is 3.
[Sc^{3+}, Ti^{3+}], [Mn^{2+}, Cr^{2+}], [Cu^{2+}, Zn^{2+}] and [Ni^{2+}, Ti^{4+}]

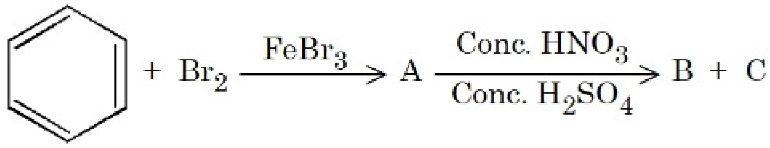
Statement II: Th^{4+} is the strongest reducing agent among Th^{4+}, Ce^{4+}, Gd^{3+} and Eu^{2+}.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options
1. Both Statement I and Statement II are false
2. Both Statement I and Statement II are true
3. Statement I is false but Statement II is true
4. Statement I is true but Statement II is false

Question Type : **MCQ**
Question ID : **444792736**
Option 1 ID : **4447922507**
Option 2 ID : **4447922506**
Option 3 ID : **4447922509**
Option 4 ID : **4447922508**

Q.56 Method used for separation of mixture of products (B and C) obtained in the following reaction is



Options
1. sublimation
2. fractional distillation
3. simple distillation
4. steam distillation

Question Type : **MCQ**
Question ID : **444792737**
Option 1 ID : **4447922510**
Option 2 ID : **4447922512**
Option 3 ID : **4447922513**
Option 4 ID : **4447922511**

Q.57 The wave numbers of three spectral lines of H atom are considered. Identify the set of spectral lines belonging to Balmer series.
(R = Rydberg constant)

Options
1. $\frac{3R}{4}, \frac{3R}{16}, \frac{7R}{144}$
2. $\frac{7R}{144}, \frac{3R}{16}, \frac{16R}{255}$
3. $\frac{5R}{36}, \frac{8R}{9}, \frac{15R}{16}$
4. $\frac{5R}{36}, \frac{3R}{16}, \frac{21R}{100}$

Question Type : **MCQ**
Question ID : **444792727**
Option 1 ID : **4447922471**
Option 2 ID : **4447922473**
Option 3 ID : **4447922470**
Option 4 ID : **4447922472**

Q.58 $Ph-CH=CH_2 \xrightarrow[HBr]{(PhCOO)_2} \text{Product}$

Consider the above reaction

A. The reaction proceeds through a more stable radical intermediate.
B. The role of peroxide is to generate $H^{\bullet}$ (Hydrogen radical).
C. During this reaction, benzene is formed as a byproduct.
D. 1−Bromo−2−phenylethane is formed as the minor product.
E. The same reaction in absence of peroxide proceeds via carbocation intermediate.

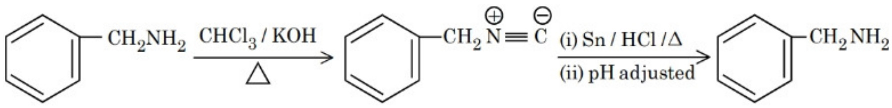
Identify the correct statements. Choose the ***correct*** answer from the options given below:

Options
1. A, C & E Only
2. A & E Only
3. A, B & D Only
4. C, D & E Only

Question Type :	**MCQ**
Question ID :	**444792739**
Option 1 ID :	**4447922520**
Option 2 ID :	**4447922518**
Option 3 ID :	**4447922519**
Option 4 ID :	**4447922521**

Q.59 Consider the following reactions giving major product. Identify the correct reaction.

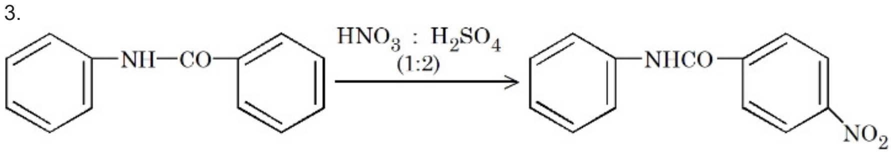
Options 1.



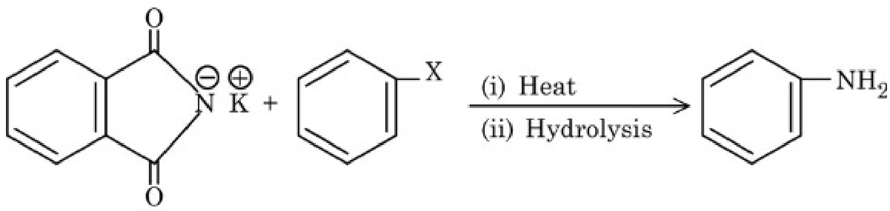
2.

$$H_3CCH_2-\overset{O}{\overset{\|}{C}}-NH_2 + Br_2 + 4\ KOH\ (alc) \xrightarrow{\Delta} H_3CCH_2NH_2 + 2KBr + K_2CO_3 + 2H_2O$$

3.

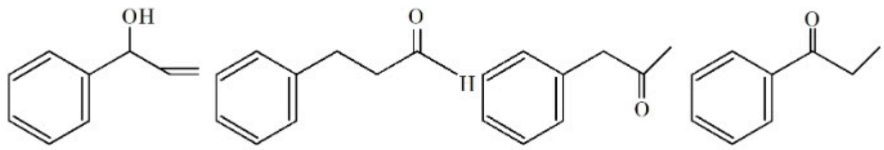


4.



Question Type :	**MCQ**
Question ID :	**444792743**
Option 1 ID :	**4447922536**
Option 2 ID :	**4447922537**
Option 3 ID :	**4447922534**
Option 4 ID :	**4447922535**

Q.60 Given below are the four isomeric compounds (P, Q, R, S)



(P) (Q) (R) (S)

Identify **correct** statements from below.

A. Q, R and S will give precipitate with 2, 4 − DNP.
B. P and Q will give positive Bayer's test.
C. Q and R will give sooty flame.
D. R and S will give yellow precipitate with I_2 / NaOH.
E. Q alone will deposit silver with Tollen's reagent

Choose the correct option.

Options
1. C and E only
2. A and E only
3. A, C and E only
4. A, B, D and E only

Question Type : **MCQ**
Question ID : **444792742**
Option 1 ID : **4447922530**
Option 2 ID : **4447922533**
Option 3 ID : **4447922531**
Option 4 ID : **4447922532**

Q.61 Regarding the hydrides of group 15 elements EH_3 (E = N, P, As, Sb), select the correct statement from the following:

A. The stability of hydrides decreases down the group.
B. The basicity of hydrides decreases down the group.
C. The reducing character increases down the group.
D. The boiling point increases down the group.

Choose the *correct* answer from the options given below:

Options
1. A, B, C & D
2. B & C only
3. A & D only
4. A, B & C only

Question Type : **MCQ**
Question ID : **444792734**
Option 1 ID : **4447922498**
Option 2 ID : **4447922499**
Option 3 ID : **4447922500**
Option 4 ID : **4447922501**

Q.62 Consider the following reaction sequence

$$\underset{\text{vapour density 47]}}{\underset{[76.6\%\ \text{C},\ 6.38\%\ \text{H},}{\text{Compound (x)}}} \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CO_2,\ NaOH,\ 120°C,\ \text{high pressure}} \underset{\text{(Major Product)}}{\text{Compound (y)}}$$

Compound (y) develops characteristic colour with neutral $FeCl_3$ solution.
Identify the **INCORRECT** statement from the following for the above sequence.

Options
1. Compound y will dissolve in $NaHCO_3$ and evolve a gas.
2. Compound x is more acidic than compound y.
3. Both compounds x and y will dissolve in NaOH.
4. Both compounds x and y will burn with sooty flame.

Question Type : **MCQ**
Question ID : **444792741**
Option 1 ID : **4447922527**
Option 2 ID : **4447922529**
Option 3 ID : **4447922526**
Option 4 ID : **4447922528**

Q.63 At T(K), 2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg. At this stage, one mole of A and one mole of B are added to the solution. The vapour pressure is now measured as 328.6 mm Hg. The vapour pressure (in mm Hg) of A and B are respectively:

Options
1. 500, 200
2. 300, 200
3. 400, 300
4. 600, 400

Question Type : **MCQ**
Question ID : **444792730**
Option 1 ID : **4447922483**
Option 2 ID : **4447922485**
Option 3 ID : **4447922482**
Option 4 ID : **4447922484**

Q.64 CORRECT order of stability for the following is

$CH_2 = CH^-$, $CH_3 - CH_2^-$, $CH \equiv C^-$

Options
1. $CH \equiv C^- > CH_3 - CH_2^- > CH_2 = CH^-$
2. $CH_3 - CH_2^- > CH_2 = CH^- > CH \equiv C^-$
3. $CH_2 = CH^- > CH \equiv C^- > CH_3 - CH_2^-$
4. $CH \equiv C^- > CH_2 = CH^- > CH_3 - CH_2^-$

Question Type : **MCQ**
Question ID : **444792738**
Option 1 ID : **4447922516**
Option 2 ID : **4447922515**
Option 3 ID : **4447922514**
Option 4 ID : **4447922517**

Q.65 Consider a weak base 'B' of $pK_b = 5.699$. 'x' mL of 0.02 M HCl and 'y' mL of 0.02 M weak base 'B' are mixed to make 100 mL of a buffer of pH 9 at 25 °C. The values of 'x' and 'y' respectively are:
[Given: log 2 = 0.3010, log 3 = 0.4771, log 5 = 0.699]

Options

1.

x	y
85.7	14.3

2.

x	y
42.7	57.3

3.

x	y
11.1	88.9

4.

x	y
14.3	85.7

Question Type : **MCQ**
Question ID : **444792731**
Option 1 ID : **4447922488**
Option 2 ID : **4447922486**
Option 3 ID : **4447922487**
Option 4 ID : **4447922489**

Q.66 Given below are two statements:

Statement I: Griss−Ilosvay test is used for the detection of nitrite ion, which involves the use of sulphanilic acid and α−naphthylamine reagent.

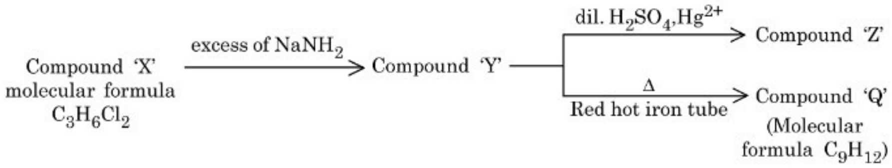
Statement II: In the above test, sulphanilic acid is diazotized by the acidified nitrite ion, which on further coupling with α−naphthylamine forms an azo−dye.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options
1. Statement I is false but Statement II is true
2. Statement I is true but Statement II is false
3. Both Statement I and Statement II are false
4. Both Statement I and Statement II are true

Question Type : **MCQ**
Question ID : **444792745**
Option 1 ID : **4447922545**
Option 2 ID : **4447922544**
Option 3 ID : **4447922543**
Option 4 ID : **4447922542**

Q.67 Given below are two statements for the following reaction sequence.



Statement I: Compound 'Z' will give yellow precipitate with NaOI.

Statement II: Compound 'Q' has two different types of 'H' atoms (aromatic : aliphatic) in the ratio 1:3.

In the light of the above statements, choose the ***correct*** answer from the options given below:

Options
1. Statement I is false but Statement II is true
2. Both Statement I and Statement II are true
3. Both Statement I and Statement II are false
4. Statement I is true but Statement II is false

Question Type : **MCQ**
Question ID : **444792740**
Option 1 ID : **4447922525**
Option 2 ID : **4447922522**
Option 3 ID : **4447922523**
Option 4 ID : **4447922524**

Q.68 The correct statement among the following is:

Options
1. $Ni(CO)_4$ is diamagnetic and $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are paramagnetic.
2. $[Ni\,(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ are diamagnetic and $Ni(CO)_4$ is paramagnetic.
3. $Ni(CO)_4$ and $[NiCl_4]^{2-}$ are diamagnetic and $[Ni(CN)_4]^{2-}$ is paramagnetic.
4. $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $[NiCl_4]^{2-}$ is paramagnetic.

Question Type : **MCQ**
Question ID : **444792735**
Option 1 ID : **4447922505**
Option 2 ID : **4447922504**
Option 3 ID : **4447922503**
Option 4 ID : **4447922502**

Q.69

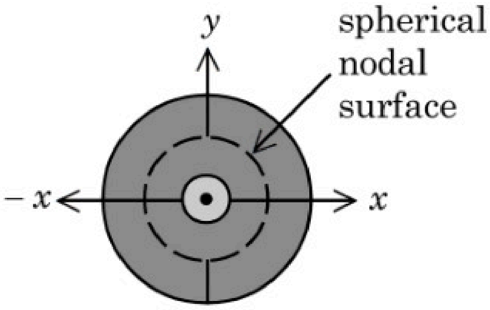


Figure 1. electron probability density for 2s orbital

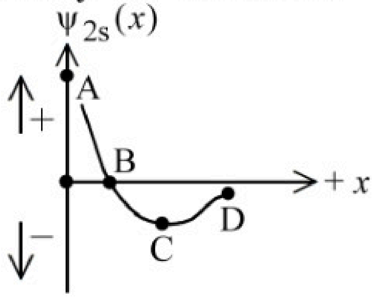


Figure 2. wave function for 2s orbital

Which of the following point in Figure 2 most accurately represents the nodal surface as shown in Figure 1?

Options
1. D
2. A
3. C
4. B

Question Type : **MCQ**
Question ID : **444792726**
Option 1 ID : **4447922469**
Option 2 ID : **4447922466**
Option 3 ID : **4447922468**
Option 4 ID : **4447922467**

Q.70 In period 4 of the periodic table, the elements with highest and lowest atomic radii are respectively.

Options
1. K & Se
2. K & Br
3. Rb & Br
4. Na & Cl

Question Type : **MCQ**
Question ID : **444792733**
Option 1 ID : **4447922497**
Option 2 ID : **4447922495**
Option 3 ID : **4447922496**
Option 4 ID : **4447922494**

Section : **Chemistry Section B**

Q.71 0.53 g of an organic compound (x) when heated with excess of nitric acid (concentrated) and then with silver nitrate gave 0.75 g of silver bromide precipitate. 1.0 g of (x) gave 1.32 g of CO_2 gas on combustion. The percentage of hydrogen in the compound (x) is ____%. [Nearest Integer]

[Given: Molar mass in g mol^{-1} H : 1, C : 12, Br : 80, Ag : 108, O : 16 ; Compound (x) : $C_xH_yBr_z$]

Question Type : **SA**
Question ID : **444792747**

Q.72 Consider the following redox reaction taking place in acidic medium

$$BH_4^-(aq) + ClO_3^-(aq) \longrightarrow H_2BO_3^-(aq) + Cl^-(aq)$$

If the Nernst equation for the above balanced reaction is

$$E_{cell} = E^{\circ}_{cell} - \frac{RT}{nF}\ln Q,$$

then the value of n is _______ .(Nearest integer)

Question Type : **SA**
Question ID : **444792749**

Q.73 X is the number of geometrical isomers exhibited by $[Pt(NH_3)(H_2O)BrCl]$.

Y is the number of optically inactive isomer(s) exhibited by $[CrCl_2(ox)_2]^{3-}$

Z is the number of geometrical isomers exhibited by $[Co(NH_3)_3(NO_2)_3]$.

The value of X + Y + Z is ______.

Question Type : **SA**
Question ID : **444792746**

Q.74 Consider the dissociation equilibrium of the following weak acid

$$HA \rightleftharpoons H^+(aq) + A^-(aq)$$

If the pKa of the acid is 4, then the pH of 10 mM HA solution is _____ .(Nearest integer)

[Given: The degree of dissociation can be neglected with respect to unity]

Question Type : **SA**
Question ID : **444792750**

Q.75 500 mL of 1.2 M KI solution is mixed with 500 mL of 0.2 M $KMnO_4$ solution in basic medium. The liberated iodine was titrated with standard 0.1 M $Na_2S_2O_3$ solution in the presence of starch indicator till the blue color disappeared. The volume (in L) of $Na_2S_2O_3$ consumed is _____. (Nearest integer)

Question Type : **SA**
Question ID : **444792748**